

(A01) ERNE Fittings GmbH Hauptstraße 48 6824 Schlins Austria		INSPECTION CERTIFICATE (A02) 3.1 EN10204:2004 No.: 2022-014943-SO (A03) Rev.: 0 Date: 2022-04-27 Page: 1 / 4
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(A06)

Customer	Consignee
Erne Fittings USA, Inc. 520 Post Oak Boulevard Suite 710 TX 77027 Houston United States	Dodson Global - Houston 2 2750 Holmes Road TX 77051 Houston United States

Order (A07, A08)

Erne order no.	000875DM
Erne order pos.	0140
Customer order no.	305423
Order date	2021-11-04

Requirements of the order (B01, B02, B03, B04)

cap MSS SP-75-14 + CSA Z245.11-17 material . . . : WPHY 60/WPHY 65/Gr. 448 Cat-II TR-fitting . . : MSS SP-75-2014 : CSA Z245.11-17 TR-addit. fitting : NACE MR0175/ISO15156-2:2015A.2 NACE MR0103/ISO17945:2015 13.1 & 15 : For WPHY65: ASTM A860-18 certificate . . : EN 10204:2004/3.1 rust protection. : Millstandard

Item specification (B01)

Manufacturer item no.	178852
Product form	cap
Article designation	CAP-S-WPHY60/65/448CatII-16"-XS
Steel grade	WPHY_60/WPHY_65/Gr._448_Cat-II

Product dimensions (nominal) (B09, B10, B11)

Diameter	406.40 mm
Wall thickness	12.70 mm

Product (B04, B07, B08)

Code No.	1DF4T
Quantity	15 pcs
Manufacturing process	Hot formed

Marking, Identification

 AUSTRIA 2245.11-17 448 M46C SS A860 WPHY60/WPHY65 QT 1DF4T 16" XS 

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Pre material
(C70)

Heat no.	Starting material
1NAU55	plate

Heat analysis
(C71)

Heat no.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	V
1NAU55	0.16 %	0.21 %	1.45 %	0.015 %	0.002 %	0.03 %	< 0.01 %	0.02 %	0.05 %	< 0.01 %

Heat no.	Nb	Al	B	Ti	N	CE _{IIW}
1NAU55	0.02 %	0.040 %	0.0003 %	< 0.01 %	0.01 %	< 0.41

Product analysis
(C81)

Heat no.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	V
1NAU55	0.178 %	0.24 %	1.39 %	0.014 %	< 0.001 %	0.03 %	0.00 %	0.03 %	0.05 %	0.00 %

Heat no.	Nb	Al	B	Ti	N	CE _{IIW}	CE _N
1NAU55	0.02 %	0.042 %	0.0001 %	0.00 %	0.019 %	0.42	0.42

Formula legend
(C71)

Code	Name	Formula
CE _{IIW}	Carbon Equivalent (IIW)	$C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$
CE _N	Carbon Equivalent (N)	$C+(0.75+0.25*\tanh(20*(C-0.12)))*(Si/24 + Mn/6 + Cu/15 + Ni/20 + (Cr+Mo+Nb+V)/5 + 5B)$

Heat treatment
(B04)

Process	Temperature	Cooling
Quenched and tempered		
	920 °C	Cooling in water
	640 °C	Cooling in still air

Hardness test
(C30, C31)

Test report no.	Method	Min	Max
15674	HB	-	190

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Tensile test

(C01, C02, C03, C10, C11, C12, C13)

Sample no.	Sample orientation	Sample size	Temperature	Yield Strength	Tensile strength	Elongation
				R _{p0.2}		
IE6_689	Transversal	Round specimen Ø 8.82 mm L: 35.60 mm 4D	+23 °C	66000 PSI	89500 PSI	27 %

Notched bar impact test

(C40, C41, C42, C43, C44)

Sample no.	Sample orientation	Temperature	Dimension of test sample			Work	Lat. width	Shear fracture
			bo	So	Notch shape			
IE6_689	Longitudinal	-46 °C	10.0 mm	80.0 mm ²	V-Notch	169.0 J 175.0 J 230.0 J Ø 191.3 J	2.37 mm 2.37 mm 2.51 mm	- - -
IE6_698	Transversal	-46 °C	10.0 mm	80.0 mm ²	V-Notch	187.0 J 231.0 J 228.0 J Ø 215.3 J	- - -	95.0 % 95.0 % 95.0 %

Additional tests

(D02, D51)

Test name	Test report no.	Test standard	Conditions	Rate	Result
Visual inspection and dimensional check					Without objection

Statement of Compliance

(Z01)

The products were manufactured, sampled, tested and inspected in accordance with the specifications and were found to meet the requirements.

Inspection representative

(A05, Z02, Z03)

Date	2022-04-27
Validated by	Inspection Representative
Name / Signature / Stamp	Sommer Stefanie
Telephone	+43 5524 501 328
Mail	stefanie.sommer@ernefittings.com

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Indication in parentheses correspond to attributes according to EN 10168

Official approvals

(Z05)

Approved by	Name	Certificate No.
TÜV SÜD Industrie Service GmbH	AD 2000 - Merkblatt W0, Pressure Equipment Directive (PED) 2014/68/EU Annex I, Paragraph 4.3 ; EN 764-5, Para. 4.2	DGR-0036-QS-W 129/2002/MUC-01
TÜV SÜD Industrie Service GmbH	EN ISO 9001:2015	12 100 3054 TMS

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